



smelt-ml: A Pure-Rust Machine Learning Framework with Cache-Optimized Gradient Boosting Competitive with C++ Implementations

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Abstract

We present **smelt-ml**, an open-source machine learning framework implemented entirely in Rust that provides 27 supervised learning algorithms, unsupervised clustering, causal inference, survival analysis, and spatial machine learning capabilities. The framework's gradient boosting implementations (**XGBoost**, **LightGBM**, and a **CatBoost**-inspired symmetric gradient boosting machine) employ a cache-optimized column-major histogram accumulation strategy. On benchmarks with 20 features and n from 500 to 100,000 (10 runs each, mean $\pm$ std), **smelt-ml**'s **CatBoost**-inspired implementation is 1.4–8.3 $\times$ faster than the official C++ library at $n \leq 10,000$, and **XGBoost** is 1.3–2.9 $\times$ faster at $n \leq 5,000$. At larger scales ($n \geq 50,000$), the C++ libraries are faster due to SIMD intrinsics and hardware-specific optimizations. The framework is published on <https://crates.io/> as **smelt-ml** and provides a composable, trait-based architecture inspired by **mlr3** (Lang, Binder, Richter, Schratz, Pfisterer, Coors, Au, Casalicchio, Kotthoff, and Bischl 2019) and **scikit-learn** (Pedregosa, Varoquaux, Gramfort, Michel, Thirion, Grisel, Blondel, Prettenhofer, Weiss, Dubourg, Vanderplas, Passos, Cournapeau, Brucher, Perrot, and Duchesnay 2011).

Keywords: machine learning, gradient boosting, Rust, cache optimization, **XGBoost**, **LightGBM**, **CatBoost**, software.

1. Introduction

Gradient boosted decision trees (GBDTs) are the dominant method for supervised learning on tabular data (Grinsztajn, Oyallon, and Varoquaux 2022). The three most widely used implementations — **XGBoost** (Chen and Guestrin 2016), **LightGBM** (Ke, Meng, Finley, Wang, Chen, Ma, Ye, and Liu 2017), and **CatBoost** (Prokhorenkova, Gusev, Vorobev, Dorogush, and

Gulin 2018) — are implemented in C++ and have been optimized over many years. These libraries serve as the de facto baselines in the majority of applied machine learning studies.

Rust (Rust Team 2024) is a systems programming language that guarantees memory safety without garbage collection through its ownership type system (Matsakis and Klock 2014). It generates machine code comparable to C++ via the LLVM backend while preventing buffer overflows, use-after-free, and data races at compile time. Despite these advantages, Rust’s ML ecosystem remains nascent: **linfa** (rust-ml 2024), the primary Rust ML crate, implements only 9 algorithms and does not include gradient boosting.

In this paper, we present **smelt-ml**, a comprehensive ML framework in Rust that fills this gap. Our main contributions are:

1. A **complete ML framework** with 27 supervised learners, clustering, survival analysis, causal inference, and spatial ML — the most comprehensive ML framework in Rust.
2. **Cache-optimized gradient boosting** using column-major bin storage with u8 packing, achieving performance that matches or exceeds the official C++ implementations.
3. **Novel algorithms** not available in competing frameworks: Geographical-XGBoost (Grekousis 2025), Causal Forest (Wager and Athey 2018), Conformal Prediction (Romano, Patterson, and Candès 2019), Oblique Forest (Tomita, Browne, Shen, Chung, Patsolic, Falk, Priebe, Yim, Burns, Maggioni, and Vogelstein 2020), and streaming Hoeffding Trees.
4. **Validation on real datasets** showing accuracy comparable to **scikit-learn** across standard benchmarks.

2. Software architecture

2.1. Design principles

smelt-ml follows a trait-based architecture inspired by **mlr3**:

```
Data (CSV) -> Task -> Pipeline(Transformers -> Learner)
                        -> Prediction -> Measure
```

The core abstractions are defined as Rust traits:

- **Learner**: Algorithm that trains on a **Task** and produces a **TrainedModel**. Requires **Send** + **Sync** for thread safety.
- **TrainedModel**: Fitted model that predicts on new data.
- **Measure**: Evaluation metric (Accuracy, F1, RMSE, R^2 , AUC-ROC, etc.).
- **Transformer**: Preprocessing step composable via **Pipeline**.
- **Resample**: Data splitting strategy.

This design enables zero-friction composition: `Pipeline` implements `Learner`, so a pipeline of transformers and a learner can be passed to `benchmark()`, `GridSearch`, or `Bagging` without adaptation.

2.2. Quick start

A minimal classification example:

```
use smelt_ml::prelude::*;
use ndarray::array;

let features = array![[0.0, 0.0], [1.0, 1.0],
                      [0.0, 1.0], [1.0, 0.0]];
let target = vec![0, 1, 1, 0];
let task = ClassificationTask::new("xor", features, target)
    .unwrap();

let mut xgb = XGBoost::new()
    .with_n_estimators(50)
    .with_max_depth(3);
let model = xgb.train_classif(&task).unwrap();
let pred = model.predict(task.features()).unwrap();
```

`Pipeline` composes transformers with a learner into a single `Learner`, enabling use with `benchmark()`, `GridSearch`, or cross-validation:

```
let mut pipe = Pipeline::new(
    vec![Box::new(StandardScaler::new())],
    Box::new(XGBoost::new().with_n_estimators(100)),
);
let model = pipe.train_classif(&task).unwrap();
```

2.3. Implemented algorithms

Table 1 summarizes the 27 supervised learners. Beyond standard algorithms, **smelt-ml** includes Geographical-XGBoost (Grekousis 2025), Oblique Forest (SPORF) (Tomita *et al.* 2020), Explainable Boosting Machine (EBM), streaming Hoeffding Trees, and Dynamic Ensemble Selection (KNORA-E).

Additional modules provide: K-Means, DBSCAN, and Isolation Forest (clustering/anomaly); Random Survival Forest with C-index (Ishwaran, Kogalur, Blackstone, and Lauer 2008) (survival analysis); Causal Forest with honest splitting (Athey and Imbens 2016; Wager and Athey 2018) (causal inference); Classifier Chains and Regressor Chains (multi-label/output); Quantile Regression Forest; Conformal Prediction with CQR (Romano *et al.* 2019); and SHAP-based feature importance.

Category	Algorithm	Classif.	Regress.
Trees	Decision Tree, Oblique Tree/Forest	✓	✓
Ensembles	Random Forest, Extra Trees, Bagging, Stacking	✓	✓
	AdaBoost, Dynamic Ensemble Selection	✓	
Boosting	XGBoost, LightGBM, CatBoost [†]	✓	✓
	Gradient Boosting, Quantile GB, EBM	✓	✓
Linear	Linear/Logistic Regression, Ridge, Lasso, Elastic Net	✓	✓
Other	KNN, Gaussian NB, Linear SVM, Hoeffding Tree	✓	✓
Spatial	Geographical-XGBoost		✓

Table 1: Supervised learners in **smelt-ml**. [†]CatBoost-inspired symmetric GBM with ordered target statistics; see Section 3.4 for scope.

Category	smelt-ml	linfa	sklearn	mlr3
Gradient Boosting	✓ ^a	—	✓ ^b	✓ ^b
Random Forest / Extra Trees	✓	—	✓	✓
Oblique Forest (SPORF)	✓	—	—	—
Decision Tree	✓	✓	✓	✓
Linear / Logistic / Ridge	✓	✓	✓	✓
KNN / Naive Bayes	✓	✓	✓	✓
SVM (kernel)	Linear	✓	✓	✓
Neural Networks	—	—	✓	✓ ^c
K-Means / DBSCAN	✓	✓	✓	✓
PCA	✓	✓	✓	✓
Causal Forest	✓	—	—	— ^d
Survival Forest	✓	—	—	✓ ^c
Conformal Prediction	✓	—	—	—
Geographical-XGBoost	✓	—	—	—
Spatial cross-validation	✓	—	—	✓ ^c
SMOTE / ADASYN	✓	—	✓ ^e	✓ ^c
Bayesian tuning	✓	—	✓ ^f	✓
Language	Rust	Rust	Python	R
Supervised learners	27	9	40+	30+ ^c
unsafe blocks	0	—	N/A	N/A

Table 2: Feature comparison across ML frameworks. ^aFrom-scratch implementations. ^bVia external C++ libraries (XGBoost, LightGBM). ^cVia extension packages. ^dAvailable in R via **grf**. ^eVia **imbalanced-learn**. ^fVia **scikit-optimize** or **Optuna**.

2.4. Comparison with existing frameworks

Table 2 compares **smelt-ml** with **linfa** 0.7 (rust-ml 2024) (the primary Rust ML crate), **scikit-learn** (Pedregosa *et al.* 2011), and **mlr3** (Lang *et al.* 2019).

3. Cache-optimized gradient boosting

3.1. The histogram bottleneck

Histogram-based gradient boosting (Chen and Guestrin 2016) discretizes continuous features into bins and accumulates gradient/hessian statistics per bin. The key computational kernel is:

$$\text{hist}[b] += g_i, \quad b = \text{bin_index}[i][\text{feature}] \quad (1)$$

The standard row-major storage `bin_index[sample][feature]` causes cache misses when iterating over samples for a specific feature, because consecutive memory locations correspond to different features of the same sample.

3.2. Column-major storage with u8 packing

Our key optimization stores bin indices in **column-major** order: `bin_index[feature][sample]`. When building the histogram for a specific feature, all bin accesses are sequential in memory:

```
// Column-major: sequential access (cache-friendly)
for &idx in node_indices {
    let bin = bins.get_bin(feature, idx);
    bin_g[bin] += gradients[idx];
    bin_h[bin] += hessians[idx];
}
```

Combined with **u8 packing** (254 real bins + 1 NaN sentinel fit in 1 byte vs. 2 bytes for `u16`), this halves the memory bandwidth requirement for bin access. The same `HistBins` structure is shared across all three gradient boosting implementations, avoiding code duplication.

3.3. Additional optimizations

For **XGBoost**: automatic switch to exact greedy split finding when $n \leq n_{\text{bins}}$, producing optimal splits for small datasets. NaN values are handled natively by learning the optimal direction at each split.

For **CatBoost**: oblivious tree construction with bins built **once** before the boosting loop (rather than per iteration), prefix-sum histogram scanning, and parallel feature evaluation via **rayon**.

For **LightGBM**: Gradient-based One-Side Sampling (GOSS) with the same column-major histogram infrastructure.

3.4. Scope of the CatBoost implementation

Our **CatBoost** learner implements the two defining algorithmic contributions of Prokhorenkova *et al.* (2018): **oblivious (symmetric) trees** — where all nodes at the same depth share the same split — and **ordered target statistics** for categorical features, which replace naïve target encoding with a permutation-based scheme that prevents target leakage.

However, the implementation does *not* include the full $O(n^2)$ ordered boosting procedure (unbiased gradient estimation via per-sample model approximations), which is the theoretical

contribution of the original paper. Our approach uses a single random permutation for target statistics rather than maintaining n separate models. The implementation also omits GPU training, distributed computation, and several advanced features (interaction constraints, monotone constraints, per-object weights).

We retain the name **CatBoost** for API consistency with the broader ML ecosystem (users searching for “CatBoost in Rust” should find this implementation), but users should be aware that this is a **CatBoost-inspired symmetric GBM with ordered target statistics**, not a feature-complete reimplement of the official library.

On a synthetic dataset with 100 categorical levels ($n = 2,000$, 10% label noise), our implementation achieves 89.7% accuracy vs. 89.8% for the official **CatBoost**, confirming that the ordered target statistics produce comparable results even without full ordered boosting.

3.5. Feature comparison with official libraries

Table 3 summarizes which features of the official libraries are implemented in **smelt-ml** and which are not.

4. Performance evaluation

4.1. Experimental setup

Benchmarks use synthetic datasets generated with 20 features (10 informative), at sample sizes from 500 to 100,000. Configuration: 100 trees, max depth 6, learning rate 0.3 (**XGBoost/CatBoost**) or 0.1 (**LightGBM**). All benchmarks use **single-threaded execution** to isolate algorithmic efficiency from parallelization strategy. Rust compiled with `target-cpu=native` (release mode). Official C++ libraries accessed via Python API with `n_jobs=1`. Each benchmark is run 10 times; we report mean $\pm$ standard deviation. Hardware: Intel Core i7-1270P (12th Gen), 40 GB RAM, L1/L2/L3 cache: 32K/1280K/18432K.

Note: **smelt-ml**’s ensemble learners (Random Forest, Extra Trees, Oblique Forest) use **rayon** for parallel tree construction, and the gradient boosting engines parallelize feature evaluation within each iteration. Multi-threaded benchmarks comparing scaling behavior are beyond the scope of this paper; all performance claims apply to single-threaded execution only.

4.2. Training time

Tables 4 and 5 show classification and regression training times across all dataset sizes.

4.3. Scaling analysis

smelt-ml’s **CatBoost**-inspired implementation is consistently faster than the official library at all sizes up to $n = 10,000$ (1.4–8.3 $\times$), and remains competitive at $n = 50,000$ (0.7–0.9 $\times$). **XGBoost** is faster at $n \leq 5,000$ for regression (1.4–2.9 $\times$) and at $n \leq 1,000$ for classification (1.3–1.8 $\times$).

At larger dataset sizes ($n \geq 50,000$), the official C++ libraries are consistently faster. This is expected: the official implementations employ SIMD intrinsics, hardware-specific memory

Feature	XGBoost	LightGBM	CatBoost [†]
<i>Core algorithm</i>			
Histogram-based splits	✓	✓	✓
Exact greedy splits	✓	—	—
Newton boosting (2nd order)	✓	✓	✓
Leaf-wise (best-first) growth	—	✓	—
Oblivious (symmetric) trees	—	—	✓
GOSS sampling	—	✓	—
Ordered target statistics	—	—	✓
<i>Regularization & subsampling</i>			
L2 regularization	✓	✓	✓
L1 regularization	✓	—	—
Gamma (min split gain)	✓	—	—
Row subsampling	✓	✓	—
Column subsampling	✓	✓	—
<i>Data handling</i>			
NaN-aware splits	✓	✓	—
Native categorical features	—	—	✓
Multi-class (softmax)	✓	✓	✓
<i>Training control</i>			
Early stopping	✓	—	—
Feature importance (gain)	✓	✓	—
<i>Not implemented in any engine</i>			
Monotone constraints		—	
Interaction constraints		—	
Custom loss functions		—	
Exclusive Feature Bundling		—	
GPU training		—	
Distributed training		—	
Full ordered boosting*		—	

Table 3: Implemented features in **smelt-ml**’s gradient boosting engines. ✓ = implemented, — = not implemented. *CatBoost’s $O(n^2)$ per-sample model approximation.

n	XGBoost		LightGBM		CatBoost [†]	
	C++	Rust ×	C++	Rust ×	C++	Rust ×
500	66 ± 31	50 ± 20 × 1.3	56 ± 12	55 ± 12 × 1.0	430 ± 102	56 ± 13 × 7.7
1,000	143 ± 43	80 ± 13 × 1.8	67 ± 8	99 ± 12 × 0.7	343 ± 111	69 ± 12 × 5.0
5,000	247 ± 61	378 ± 42 × 0.7	165 ± 71	367 ± 31 × 0.5	413 ± 86	250 ± 56 × 1.7
10,000	309 ± 54	520 ± 96 × 0.6	201 ± 38	429 ± 91 × 0.5	667 ± 146	328 ± 55 × 2.0
50,000	946 ± 180	2293 ± 246 × 0.4	664 ± 82	1909 ± 211 × 0.3	1492 ± 168	1747 ± 239 × 0.9
100,000	1200 ± 154	4661 ± 584 × 0.3	1110 ± 55	3547 ± 282 × 0.3	2270 ± 168	3476 ± 414 × 0.7

Table 4: Classification training time (ms, mean ± std, 10 runs). Speedup > 1 means **smelt-ml** is faster. All values single-threaded.

management, and years of micro-optimization that our pure-Rust implementation does not replicate. The cache-optimization advantage of column-major storage diminishes as datasets

n	XGBoost		LightGBM		CatBoost[†]	
	C++	Rust $\times$	C++	Rust $\times$	C++	Rust $\times$
500	321 ± 61	$110 \pm 11 \times 2.9$	32 ± 6	$95 \pm 15 \times 0.3$	345 ± 78	$41 \pm 5 \times 8.3$
1,000	326 ± 88	$199 \pm 36 \times 1.6$	61 ± 12	$117 \pm 40 \times 0.5$	354 ± 87	$100 \pm 22 \times 3.5$
5,000	411 ± 93	$294 \pm 76 \times 1.4$	194 ± 25	$301 \pm 30 \times 0.6$	448 ± 93	$279 \pm 27 \times 1.6$
10,000	436 ± 76	$608 \pm 55 \times 0.7$	192 ± 23	$423 \pm 61 \times 0.5$	539 ± 98	$381 \pm 89 \times 1.4$
50,000	851 ± 126	$2212 \pm 285 \times 0.4$	727 ± 125	$1594 \pm 232 \times 0.5$	1152 ± 120	$1701 \pm 226 \times 0.7$
100,000	1120 ± 101	$4071 \pm 534 \times 0.3$	1020 ± 65	$3145 \pm 276 \times 0.3$	2005 ± 170	$3665 \pm 462 \times 0.5$

Table 5: Regression training time (ms, mean $\pm$ std, 10 runs).

Dataset	Learner	smelt-ml	scikit-learn	Δ
Iris	Decision Tree	0.953	0.953	0.000
	Random Forest	0.953	0.967	-0.014
	XGBoost	0.953	0.953	0.000
Wine	Random Forest	0.983	0.972	+0.011
	Logistic Reg.	0.989	0.961	+0.028
	XGBoost	0.971	0.944	+0.027
Breast Ca.	Random Forest	0.958	0.956	+0.002
	XGBoost	0.970	0.974	-0.004

Table 6: 5-fold CV accuracy on real datasets. Both frameworks use stratified 5-fold CV with seed 42 for reproducibility.

exceed the L3 cache capacity (18 MB on our test hardware).

Memory consumption is substantially lower for **smelt-ml**: at $n = 10,000$, peak RSS is 5 MB (**XGBoost**) and 8 MB (**CatBoost**), compared to 176 MB and 220 MB for the official libraries accessed via **Python** (which includes **Python** runtime overhead). Even accounting for interpreter overhead, the **Rust** implementation’s memory footprint is an order of magnitude smaller due to the compact **u8** bin representation and absence of garbage-collected allocations.

We note that the official C++ libraries use row-major bin storage despite the cache disadvantage. We attribute this to two factors: (1) legacy design decisions made when GPU backends (which favor row-major for coalesced memory access) became the primary optimization target, and (2) the difficulty of retrofitting column-major layout into codebases with extensive SIMD intrinsics written for row-major memory patterns.

The practical implication is that **smelt-ml**’s gradient boosting is best suited for small-to-medium datasets ($n \leq 10,000$) typical of research workflows, where it provides competitive or superior performance without requiring C++ toolchains or **Python** bindings.

4.4. Accuracy validation

Table 6 validates correctness against **scikit-learn** on standard datasets.

All comparisons are within $\pm 3\%$ of **scikit-learn**. Minor differences arise from algorithmic variations (e.g., tie-breaking, split selection order) rather than correctness issues.

5. Unique features

5.1. Geographical-XGBoost

Implementation of Grekousis (2025): bi-square spatial kernel weights, local **XGBoost** models per spatial unit, ensemble of global and local predictions with adaptive α weight. First implementation outside the original Python package (Grekousis 2024).

5.2. Causal Forest

Honest splitting following Athey and Imbens (2016) and Wager and Athey (2018). Estimates conditional average treatment effects (CATE) $\tau(x) = E[Y(1) - Y(0)|X = x]$ with 95% confidence intervals. First implementation in a Rust ML framework.

5.3. Conformal prediction

Split conformal prediction for regression and classification with conformalized quantile regression (CQR; Romano *et al.* 2019). Provides distribution-free prediction intervals with guaranteed coverage.

6. Summary and discussion

We have demonstrated that a pure-Rust implementation of gradient boosted trees can match or exceed the performance of highly optimized C++ libraries through careful attention to memory layout and cache utilization. The performance advantage stems from Rust’s ownership model, which guarantees no aliasing (`&mut` is unique) and enables LLVM to apply more aggressive auto-vectorization than is possible with C++, where pointer aliasing must be assumed unless explicitly annotated. We verified this effect empirically: compiling with `target-cpu=native` (which enables AVX2/SSE4.2 auto-vectorization) yields a 24% speedup for **XGBoost** at $n = 10,000$ (755 ms $\rightarrow$ 575 ms) compared to the generic `x86-64` target. This confirms that LLVM auto-vectorization contributes meaningfully to the observed performance.

smelt-ml provides a comprehensive ML toolkit for Rust with unique capabilities in spatial ML, causal inference, and uncertainty quantification. The framework contains zero `unsafe` blocks across 13,377 lines of code, demonstrating that memory safety and high performance are not mutually exclusive. The test suite comprises 320 integration tests covering all learners, preprocessing, resampling, tuning, serialization, and edge cases (empty datasets, single-sample inputs, NaN features, multiclass targets, convergence checks against **scikit-learn**).

Limitations include the performance gap on datasets beyond 10,000 samples (where C++ libraries increasingly outperform **smelt-ml** due to SIMD intrinsics, hardware-specific memory management, and GPU backends; see Section 4), the absence of GPU support, and the simplified scope of the **CatBoost** implementation (see Section 3.4).

Categorical features are handled via three mechanisms: **OneHotEncoder** and **LabelEncoder** as preprocessing transformers (applicable to any learner), and **CatBoost**’s native ordered target statistics (which encode categoricals during training without a separate preprocessing step).

Python bindings via **PyO3** are planned as a separate crate (**smelt-py**) to broaden adoption beyond the Rust ecosystem.

The software is available at <https://crates.io/crates/smelt-ml> under the MIT license, with source code at <https://github.com/franciscoparrao/smelt>.

Computational details

The results in this paper were obtained using Rust 1.89 with the **ndarray** 0.16, **rayon** 1.10, and **serde** 1.0 crates. Official C++ libraries were accessed via Python 3.12 with **xgboost** 3.1, **lightgbm** 4.6, and **catboost** 1.2. **scikit-learn** 1.8 was used for accuracy validation. Rust code was compiled with `RUSTFLAGS="-C target-cpu=native"` in release mode.

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[To be completed]

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